PHYS 210 Electricity and Magnetism

Celebration of Knowledge!

Supplementary Practice 6 - CK #2 preparation

Name: _____ Date: _____

Time: _____ Marks: _____ Practice paper - not an official exam

Attempt all questions. Supporting practice; this topic mix is not the announced main-test format.

Write all rough calculations and final answers on separate paper. Show your working.

Original source numbering is retained inside each question. Answer the entire cropped question.

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1. [____] Tutorial 4 - Question 5

5. Charge of uniform density (80 nC/m^3) is distributed throughout a hollow cylindrical region formed by two coaxial cylindrical surfaces of radii 1.0 mm and 3.0 mm. Determine the magnitude of the electric field at a point which is 2.0 mm from the symmetry axis.
   a. 7.9 N/C
   b. 9.0 N/C
   c. 5.9 N/C
   d. 6.8 N/C
   e. 18 N/C

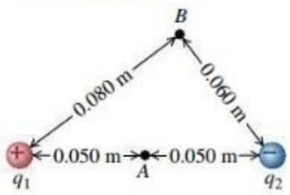
2. [____] Homework 5 - Question 23.8

23.8 •• Three equal 1.20-μC point charges are placed at the corners of an equilateral triangle with sides 0.400 m long. What is the potential energy of the system? (Take as zero the potential energy of the three charges when they are infinitely far apart.)

3. [____] Homework 5 - Question 23.19

23.19 • Two point charges $q_1 = +2.40$ nC and $q_2 = -6.50$ nC are 0.100 m apart. Point A is midway between them; point B is 0.080 m from q_1 and 0.060 m from q_2 (**Fig. E23.19**). Take the electric potential to be zero at infinity. Find (a) the potential at point A; (b) the potential at point B; (c) the work done by the electric field on a charge of 2.50 nC that travels from point B to point A.

Figure **E23.19**



4. [____] Homework 5 - Question 23.27

23.27 •• A uniformly charged, thin ring has radius 15.0 cm and total charge +24.0 nC. An electron is placed on the ring's axis a distance 30.0 cm from the center of the ring and is constrained to stay on the axis of the ring. The electron is then released from rest. (a) Describe the subsequent motion of the electron. (b) Find the speed of the electron when it reaches the center of the ring.